

HEALTH-KEY ULTRA: Publishing-Ready Content

This document contains the exact copy and metadata required for publishing the HEALTH-KEY ULTRA research across all six target platforms.

1. Zenodo (CERN) & Academia.edu

Title: HEALTH-KEY ULTRA: Repurposing Standardized Hardware Interfaces for High-Fidelity Biometric Acquisition **Authors:** Srikanth Patchava (Embedded Operating Systems Research Foundation) **Abstract / Description:** This paper presents the HEALTH-KEY ULTRA, a novel ultra-compact wearable health monitor that utilizes a dual-purpose USB Type-C interface. By implementing an active switching matrix, the device repurposes the grounded metal shield of the USB-C connector as a high-impedance analog input. This allows for single-lead electrocardiogram (ECG) and Blood Alcohol Content (BAC) acquisition without requiring dedicated, vulnerable sensor windows. The architecture also includes 64GB of local NAND flash storage, enumerating as a USB Mass Storage Class (MSC) device, ensuring data sovereignty and edge-computing capabilities. **Keywords/Tags:** Wearable Technology, USB-C, Biometric Sensing, Edge AI, Open-Source Hardware, ECG, BAC. **Files to Upload:** HEALTH_KEY_ULTRA_Scholarly_Article.pdf , HEALTH_KEY_Product_Line_White_Paper.pdf , HEALTH_KEY_Thesis.pdf .

2. ResearchGate

Project Title: HEALTH-KEY ULTRA: Edge-AI and Biometric Acquisition in Ultra-Compact Form Factors **Project Goal:** To develop a keychain-sized health monitor that eliminates cloud dependency through massive local storage and on-device machine learning (TinyML), while repurposing standard USB-C interfaces for clinical-grade sensor acquisition. **Updates/Log:** “We have successfully filed the provisional patent (US App. No.

64/073,334) detailing the USB-C shield topology. Next steps involve finalizing the TensorFlow Lite Micro models for on-device arrhythmia detection.” **Files to Upload:** HEALTH_KEY_ULTRA_Scholarly_Article.pdf (as Preprint).

3. TechRxiv (IEEE)

Title: Repurposing Standardized Hardware Interfaces for High-Fidelity Biometric Acquisition: The USB-C Shield as an Analog Front-End **Authors:** Srikanth Patchava **Categories:** Consumer Electronics, Biomedical Engineering, Circuits and Systems. **Abstract:** (Use the abstract from the Scholarly Article). **Files to Upload:** HEALTH_KEY_ULTRA_Scholarly_Article.pdf .

4. LinkedIn Article

Headline: The End of Cloud-Dependent Wearables: Introducing the HEALTH-KEY ULTRA **Body Copy:** For too long, wearable health technology has relied on a flawed premise: that your most sensitive biometric data must be continuously streamed to a third-party cloud server for processing.

At the Embedded Operating Systems Research Foundation (EoS), we are challenging this paradigm. Today, we are unveiling the research behind the **HEALTH-KEY ULTRA**, a keychain-sized device that puts data sovereignty first.

By integrating 64GB of local NAND flash and running TensorFlow Lite Micro directly on the nRF52840 MCU, the HEALTH-KEY ULTRA performs anomaly detection (like arrhythmia classification) at the edge. But the real breakthrough is physical: we’ve developed a hardware topology that repurposes the standard USB-C shield as a clinical-grade analog sensor input for ECG and breath analysis, eliminating the need for fragile sensor windows.

Read our full white paper and explore the open-source hardware repositories on our GitHub: [Link to GitHub] Read the pre-print on TechRxiv: [Link to TechRxiv]

#WearableTech #EdgeAI #OpenScience #HealthTech #EmbeddedSystems

5. Rock Health / Digital Health Wire (Press Release)

FOR IMMEDIATE RELEASE Embedded Operating Systems Research Foundation Unveils HEALTH-KEY ULTRA, Repurposing USB-C for Clinical-Grade Biometrics

[City, Date] – The Embedded Operating Systems Research Foundation (EoS), a 501©(3) nonprofit, today announced the publication of the architectural white paper for the HEALTH-KEY ULTRA, a revolutionary approach to ultra-compact health monitoring.

Unlike traditional wearables that rely on fragile optical windows and continuous cloud connectivity, the HEALTH-KEY ULTRA introduces a patented hardware topology (US Provisional App. No. ⁶⁴/_{073,334}) that utilizes the metallic shield of a standard USB Type-C connector as an active analog sensor. This allows the keychain-sized device to acquire single-lead ECG and Blood Alcohol Content (BAC) readings directly through the charging port.

“We are fundamentally rethinking how biometric data is acquired and stored,” said Srikanth Patchava, Lead Researcher at EoS. “By combining this novel sensor approach with 64GB of local storage and on-device AI inference, we are giving users complete sovereignty over their health data.”

The complete research portfolio, including open-source firmware and hardware schematics, is available via the EoS GitHub repository.

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